

Risk Aversion in Housing Prices: Effects of big earthquakes in Japan

BY ANTHONY ROCHER^a

Under the direction of SOPHIE BUHNIK^b *and* FLORENCE GOFFETTE-NAGOT^c

^a ENS de Lyon, F-69007 Lyon, France. *anthony.rocher@ens-lyon.fr*.

^b Ecole Supérieure des Professions Immobilières, 92300 Paris, France. *s.buhnik@groupe-espi.fr*.

^c Cergic, ENS de Lyon, CNRS, F-69007 Lyon, France. *florence.goffette-nagot@cnrs.fr*.

Abstract— I investigate how housing prices response to risk exposure following big earthquakes as information shocks in Japan. Using a new dataset covering whole Japan from 2015 to 2017 with ??? billions of observations, I use a DiD strategy to determine how much unaffected but still at risk areas react to the occurrence of earthquakes of intensity 6 or upper. (*JEL R30, Q54*)

How risk-averse is the housing market following a large information shock, such as a natural disaster? Standard theory suggests that, as information about risk depreciates over time, a recent shock should widen the price differential between high- and low-risk areas, conditional on an increase in the safety premium. However, the extent to which housing markets incorporate such information remains an empirical question.

Understanding how information shocks are capitalized into housing markets provides insight into whether households perceive the risks they face and how much they value newly acquired information. If the elasticity of housing prices with respect to risk exposure is close to zero, disaster-related information may have limited effects on housing markets: prices may fail to function as effective signals of underlying risk. Alternatively, households may recognize the risks but choose to remain in exposed locations, placing little or no monetary value on risk reduction. Conversely, if households fully adjust their location choices in response to new information, high-risk areas may experience declining demand and reduced attractiveness. Over time, this adjustment could lead to lower investment in exposed regions. More broadly,

persistent exposure to environmental risk may become a source of inequality, as rising safety premiums make low-risk locations increasingly inaccessible to lower-income households.

In this paper, I study how recent earthquakes affect housing prices depending on properties' exposure to seismic risk, as measured by their location on official Seismic Hazard Maps. I examine both the magnitude of the safety premium associated with lower-risk locations and how this premium changes following information shocks generated by major earthquakes.

I use a unique dataset from a Japanese real estate agency containing property listings across Japan between October 2014 and December 2017. Using detailed geocoordinates, I match each property to its corresponding level of seismic risk based on the Seismic Hazard Maps provided by Japanese disaster prevention authorities. This combination of high-frequency real estate data and spatial measures of earthquake risk allows me to examine how housing markets incorporate new information about disaster exposure.

Research question: How information shocks such as natural disasters modify risk aversion in housing prices depending on exposure to risk?

This paper contributes to the literature on hedonic pricing, initiated by Rosen (1974), and to the study of risk capitalization in housing markets. A large body of work has examined how risk perceptions are incorporated into property values, showing that differences in exposure to adverse events are reflected in equilibrium prices (Brookshire et al. (1985)).

More broadly, this paper contributes to the literature on real estate markets under uncertainty. Previous studies have investigated the impact of natural disasters on housing prices, including floods (Bin and Polasky (2004), Atreya, Ferreira, and Kriesel (2013), Atreya and Ferreira (2015)), wildfires (add references), and earthquakes (Beron et al. (1997), Sato and Shiba (2021)). These papers provide evidence that households update their valuation of properties following major risk-related events, but the extent and persistence of these adjustments remain open questions.

A growing literature has focused specifically on the effects of earthquakes in Japan, often relying on detailed administrative data provided by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT). For example, Naoi, Seko, and Sumita (2009) and Hidano, Hoshino, and Sugiura (2015) use these data to study the capitalization of earthquake risk into land and property values. This paper builds particularly on the work of Ikefuji et al. (2022), who show that land prices are [insert main finding]. I extend this line of research by examining how earthquake-related information shocks are incorporated into the broader housing market.

While previous work has investigated related questions, existing analyses have often been constrained by limited data availability. This paper leverages a newly available dataset containing billions of highly granular observations, covering both transactions and rental markets for land, houses, and apartments. The richness and accuracy of this dataset allow for a more comprehensive analysis of how housing markets respond to changes in perceived risk. The main contribution of this paper therefore lies in providing new evidence on the adjustment

of housing markets following large information shocks, exploiting an unprecedented level of detail in real estate outcomes.

My results are

The paper proceeds as follows. Section II presents a brief theory of housing prices under uncertainty. Section III presents Japan's background with earthquakes and describes the data. Section IV outlines my estimation strategy and identification assumptions. Section V presents the empirical results of the risk exposure effects on prices. Section VI uses robustness checks. Section VIII concludes.

Theoretical Framework

Let denote p the subjective probability that a property will be hit by an earthquake. p depends on both the information set i individuals possess about the seismic risks (such as a recent event, media coverage, etc.), and r the objective attributes for earthquakes to be more likely to happen, such as being located right on seismic faults. Note that π , the objective probability of an earthquake, is included in i in our case, since maps are provided and publicly available. Then the Hedonic Price Function is given by:

$$P = P(Z, r, p(i, r)),$$

with P denoting the price of the house and Z denoting an additional set of structural, environmental and locational characteristics not related to the seismic hazards maps. Following Brookshire et al. (1985), the location decision of agents is modeled using a state dependent expected utility function:

$$EU = p(i, r) \cdot U^E[Z, r, Q] + (1 - p(i, r)) \cdot U^{NE}[Z, r, Q],$$

where $U^E(.)$ is the homeowner's utility in a state in which a earthquake occurs and $U^{NE}(.)$ is the homeowner's utility when there is no such event. Q is a composite commodity, and the budget constraint of the homeowner is given by:

$$M = P(Z, r, p(i, r)) + Q$$

Maximising expected utility with respect to p subject to the budget constraint and then dividing through by the expected marginal utility of income results in the following expression:

$$\frac{\partial P}{\partial p} = \frac{U^E - U^{NE}}{p(i, r) \frac{\partial U^E}{\partial Q} + (1 - p(i, r)) \frac{\partial U^{NE}}{\partial Q}}$$

Background and Data

Japan is among the countries most exposed to seismic risk worldwide, having experienced several highly destructive earthquakes throughout its modern history. The Great Kantō Earthquake of 1923 was one of the deadliest disasters, killing more than 110,000 people and devastating large parts of Tokyo and Yokohama (Schencking (2007)). In response to the need for greater scientific understanding of seismic activity, the Earthquake Research Institute was established two years later to promote research in seismology and volcanology.

Institutional responses to earthquake risk continued to develop over the following decades. In 1981, the revision of the Building Standards Law introduced stricter seismic requirements, mandating that newly constructed buildings be designed to withstand major damage from earthquakes reaching intensity levels of 6–7 on the Japanese seismic scale. The 1995 Hanshin-Awaji Earthquake further highlighted the vulnerability of urban areas, causing extensive destruction in the Kobe region. More recently, the 2011 Tōhoku Earthquake became the most devastating earthquake in contemporary Japan, largely due to the massive tsunami that followed and significantly amplified its human and economic impacts.

Japanese people are often reminded of the earthquake threat. September 1st has been designated as Disaster Prevention Day since the Kantō Earthquake, alongside other local memorial events. The J-SHIS was created in 2005 to prevent and prepare for earthquake disasters by providing a public portal for seismic hazard information across Japan, which has been available since July 2009. These maps are based on the Seismic Hazard Map I describe below.

Over the last decades, Japan has been preparing for the so-called “megathrust earthquake,” which is believed to cause enormous damage across the country (Kobe University (2024)).

Real Estate Data

I use data from *Lifull Homes*, one of the largest real estate companies in Japan. The dataset consists of property listings covering all prefectures in Japan from January 2015 to October 2017. It contains 27,885,84 observations, including information on prices, property characteristics, and exact geolocation. Since these data come from listings, prices represent asking prices and may slightly differ from actual transaction prices. From this dataset, I retain only information on land, houses, and apartments, excluding commercial properties such as shops and other types of real estate. Table 1 describes the property types contained in the dataset.

Risk Mapping Data

The risk maps are available from the *Japan Seismic Hazard Information Station (J-SHIS)*. They have been published annually since 2009, except for 2015. Several measures of seismic risk are available. I use the so-called “Probabilistic Seismic Hazard Maps (PSHM),” which rely

Table 1: Distribution by property type

	N	Pct.
Land for sale	136,436	0.5
Houses for sale	1,372,688	4.9
Houses for rent	559,049	2.0
Appartments for sale	345,599	1.2
Appartments for rent	25,472,077	91.3
Total	27,885,849	100.0

on the concept of “exceedance probability,” referring to the probability that ground shaking exceeds a given intensity level at a specific location over a certain period (within the next 30 years). Within these measures, I use the “Average case.”¹

The maps have a high spatial resolution of 250 meters, and I match the previous dataset with these maps with high accuracy according to the listing publication year.² One may argue that these maps are affected by recent earthquakes. This is true. However, the computation process relies on a Poisson distribution based on a 100,000-year historical period, and there is limited variation from one year to another (to be checked).

Earthquake records

I use earthquake record data from the *Japan Meteorological Agency (JMA)*. The dataset includes all earthquake records from 1919 to the present, covering events that directly or indirectly affected Japanese territory. The observations include the date, location, magnitude, and intensity of each earthquake. While magnitude is the international measure based on the well-known Richter scale, I use the Japanese seismic intensity system, which measures the level of shaking experienced at a given location.³

The final selection consists of one earthquake: the 2016 Kumamoto Earthquake. This earthquake occurred on April 16th in Kyushu Prefecture, in southern Japan, and caused 277 deaths and 2,809 injuries. This earthquake sequence was the deadliest in Japan since 2011 and received extensive coverage from the Japanese media⁴.

¹A glossary is available on the J-SHIS website: <https://www.j-shis.bosai.go.jp/en/glossary>

²As the 2015 map was unfortunately the only missing year, I attribute observations from 2015 to the 2014 map.

³Intensity values are defined as “values observed using seismic intensity meters installed on the ground or on the first floor of low-rise buildings.” For example, an intensity level of 6 or higher is described as: “It is impossible to remain standing or move without crawling. People may be thrown through the air.” See JMA website for more information: <https://www.jma.go.jp/jma/en/Activities/inttable.html>

⁴citer la source.

Additional Data

To complement these data, I use Japanese census data from *e-Stat*. The census is conducted every five years, and I use the 2015 wave. The data are aggregated at the prefectural and municipal levels.⁵

Table 2: Simple regressions — Log price

	(1)	(2)	(3)	(4)	(5)
	ln_price	ln_price	ln_price	ln_price	ln_price
room_sqft	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
rooms		1.114*** (0.000)	0.781*** (0.000)	0.833*** (0.000)	0.839*** (0.000)
floor_g			0.665*** (0.000)	0.596*** (0.000)	0.593*** (0.000)
age_g				-0.414*** (0.000)	-0.416*** (0.000)
station1_distance					-0.000*** (0.000)
Constant	11.463*** (0.000)	9.748*** (0.000)	8.916*** (0.000)	9.835*** (0.001)	9.870*** (0.001)
Observations	27563098	27563098	27563098	27563095	27466272
R ²	0.000	0.405	0.567	0.610	0.611
Adj. R ²	0.000	0.405	0.567	0.610	0.611

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 1: Test of Hedonic regression

⁵The census contains information on Densely Inhabited Districts (DID), including DID population and DID area. I use these measures to compute population density as a control variable. It also provides information on the number of migrants and total population, allowing me to compute the net migration rate as a measure of local attractiveness.

Empirical Strategy

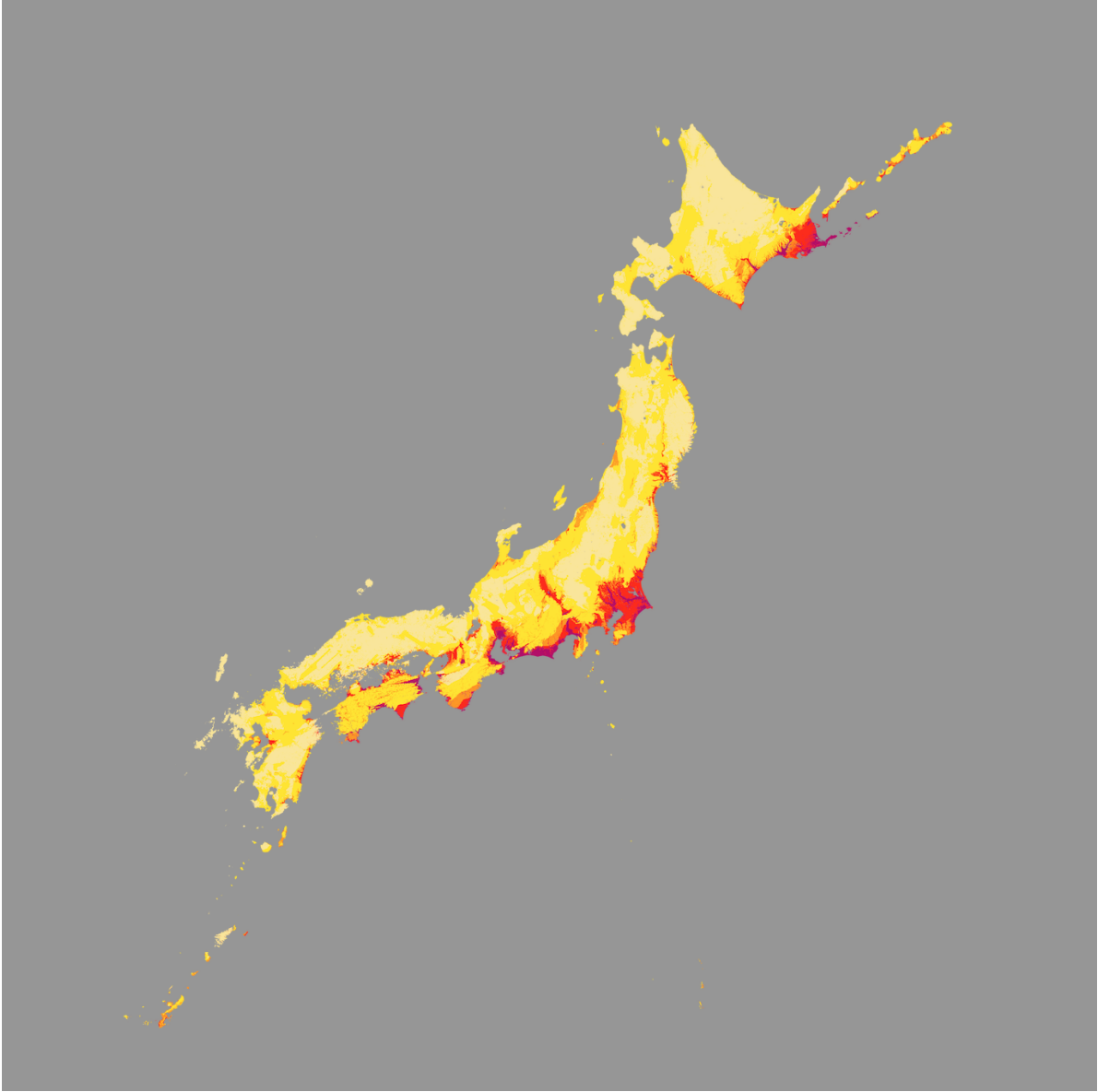
Sample selection

I focus on earthquakes occurring within Japan. This choice is motivated by an underlying “distance” hypothesis, which suggests that domestic catastrophes have a greater impact on national risk perceptions than disasters occurring abroad. This selection is also consistent with the JMA investigation zone, which records only seismic events occurring within Japanese territory.

Treatment Definition

I use Differences-in-Difference strategy. I create the treated group and the control group based on the official Seismic Hazard Maps (SHM) provided by the J-SHIS. I use the maps called “Probability of excellence” which are constructed around the probability that an earthquake will occur in the next 30 years.

To better understand what will be the control and the treated group, let’s visualize the official 2020 (the latest one) Seismic hazard risk map. The picture shows the second one show $p =$ probability that earthquake of intensity 6 + will occur in the next 30 years”.



This map is constructed from p which is the probability that an earthquake of magnitude 6 or more will strike in the next 30 years. I choose the thresholds according to the official documentation provided by the J-SHIS.

Equation

$$\ln P_{it} = \alpha_0 + \sum_{j=1}^J \alpha_j Z_{ij} + \beta r_i + \gamma Exposed_i + \delta Post_{it} + \theta_t (Post_{it} \times Exposed_i) + \varepsilon_{it}$$

Table 3: Exposure 6+ by region

	N	Pct.
Hokkaido		
Not Exposed	1,148,686	99.8
Exposed	729	0.1
Discarded	1,546	0.1
Total	1,150,961	100.0
Tohoku		
Not Exposed	202,674	92.2
Exposed	206	0.1
Discarded	17,034	7.7
Total	219,914	100.0
Kanto		
Not Exposed	3,360,174	22.1
Exposed	3,234,312	21.3
Discarded	8,588,567	56.6
Total	15,183,053	100.0
Chubu		
Not Exposed	713,151	27.5
Exposed	967,794	37.3
Discarded	911,349	35.2
Total	2,592,294	100.0
Kansai		
Not Exposed	3,839,428	60.5
Exposed	85,957	1.4
Discarded	2,420,504	38.1
Total	6,345,889	100.0
Chugoku		
Not Exposed	597,870	65.4
Exposed	1	0.0
Discarded	316,463	34.6
Total	914,334	100.0
Shikoku		
Not Exposed	32,121	7.8
Exposed	85,829	20.9
Discarded	293,399	71.3
Total	411,349	100.0
Kyushu & Okinawa		
Not Exposed	989,286	92.6
Exposed	80	0.0
Discarded	78,689	7.4
Total	9 1,068,055	100.0
Total		
Not Exposed	10,883,390	39.0
Exposed	4,374,908	15.7
Discarded	12,627,551	45.3
Total	27,885,849	100.0

P_{it} is the logarithm of the price of add i at time t , Z is a set of structural characteristics of the good. $Post_i$ takes 1 when the advertising is publised within 1 month after a big earthquake occurs.

I obtain the following results if I apply this equation to the data.

Table 4: Exposure at intensity 6+ on all kind of prices

Exposed	-0.0358*** (0.00314)
Post	0 (.)
Exposed $\times$ Post	0 (.)
Observations	575,015
R-squared	0.584
Controls	
Month $\times$ Region FE	
Standard errors in parentheses	
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

I have a problem when it comes to the *Post* variable, with Stata saying too much co-linearity with the FE. Note that the FE are here *region $\times$ year_month*. I tried as a first attempt with only *region $\times$ year*, but the same error made me opt for *region $\times$ year_month*.

With the same error, I have tried for each type of properties:

Table 5: Exposure at intensity 6+ on house sales

Exposed	0.0106* (0.00572)
Post	0 (.)
Exposed $\times$ Post	0 (.)
Observations	19,541
R-squared	0.447
Controls	
Month $\times$ Region FE	
Standard errors in parentheses	
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Controls used

- Number of rooms (`rooms`)
- Average room size (`room_sqft`)
- Floor (`floor`)
- `age_g`
 - Age < 3
 - After the Tohoku earthquake (from 2011)
 - After the Kobe earthquake (1995–2010)
 - After the para-seismic law (1981–1994)
 - After the City Planning Law (1968–1980)
 - After the WW2 (1945–1967)
- Distance to the nearest station as a proxy for distance to the center (`station1_distance`)
- Number of inhabitants of the municipality as a (bad) proxy for density (`pop_total_linear`)
 - To be replaced by number of inhabitants divided by municipality size
- Density measured as follows:
 - Rural if DID (Density Inhabited Districts) does not exist
 - $\log(\text{density})$ otherwise
- Share of migration at the municipality level as a proxy for attractiveness (`net_migration_rate`)

Robustness checks

Keeping only a few regions and apply fixed effects at the prefecture level?

Conclusion

Controls that would be nice to add:

- Average wages as a proxy for area dynamics
- Height above the sea level
- distance to the sea

Caveats:

- No control for other types of risk

Acknowledgments

Questions

References

- Atreya, Ajita, and Susana Ferreira. 2015. "Seeing Is Believing? Evidence from Property Prices in Inundated Areas." *Risk Analysis* 35 (5): 828–48. <https://doi.org/10.1111/risa.12307>.
- Atreya, Ajita, Susana Ferreira, and Warren Kriesel. 2013. "Forgetting the Flood? An Analysis of the Flood Risk Discount over Time." *Land Economics* 89 (4): 577–96. <http://www.jstor.org/stable/24243692>.
- Beron, Kurt J., James C. Murdoch, Mark A. Thayer, and Wim P. M. Vijverberg. 1997. "An Analysis of the Housing Market Before and After the 1989 Loma Prieta Earthquake." *Land Economics* 73 (1): 101–13. <https://doi.org/10.2307/3147080>.
- Bin, Okmyung, and Stephen Polasky. 2004. "Effects of Flood Hazards on Property Values: Evidence Before and After Hurricane Floyd." *Land Economics* 80 (4): 490–500. <https://doi.org/10.2307/3655805>.
- Brookshire, David S., Mark A. Thayer, John Tschirhart, and William D. Schulze. 1985. "A Test of the Expected Utility Model: Evidence from Earthquake Risks." *Journal of Political Economy* 93 (2): 369–89. <http://www.jstor.org/stable/1832182>.

- Hidano, Noboru, Tadao Hoshino, and Ayako Sugiura. 2015. "The Effect of Seismic Hazard Risk Information on Property Prices: Evidence from a Spatial Regression Discontinuity Design." *Regional Science and Urban Economics* 53 (July): 113–22. <https://doi.org/10.1016/j.regsciurbeco.2015.05.005>.
- Ikefuji, Masako, Roger J. A. Laeven, Jan R. Magnus, and Yuan Yue. 2022. "Earthquake Risk Embedded in Property Prices: Evidence from Five Japanese Cities." *Journal of the American Statistical Association* 117 (537): 82–93. <https://doi.org/10.1080/01621459.2021.1928512>.
- Kobe University. 2024. "Passing down Experiences from the Earthquake: The Kobe University of Mercantile Marine and the Great Hanshin-Awaji Earthquake." Kobe University. September 13, 2024. <https://www.kobe-u.ac.jp/en/news/article/feature0008/>.
- Naoi, Michio, Miki Seko, and Kazuto Sumita. 2009. "Earthquake Risk and Housing Prices in Japan: Evidence Before and After Massive Earthquakes." *Regional Science and Urban Economics* 39 (6): 658–69. <https://doi.org/10.1016/j.regsciurbeco.2009.08.002>.
- Rosen, Sherwin. 1974. "Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition." *Journal of Political Economy* 82 (1): 34–55. <http://www.jstor.org/stable/1830899>.
- Sato, Yasuhiro, and Keita Shiba. 2021. "The Impact of Tsunamis on Land Appraisals: Evidence from Western Japan." *PLOS ONE* 16 (4): e0248860. <https://doi.org/10.1371/journal.pone.0248860>.
- Schencking, J. Charles. 2007. "The Great Kantō Earthquake of 1923 and the Japanese Nation." Association for Asian Studies. September 2007. <https://www.asianstudies.org/publications/ea/archives/the-great-kanto-earthquake-of-1923-and-the-japanese-nation/>.